

THERESA STADLER

Data Privacy and Trustworthy AI

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👤 Personal Web

🎓 GoogleScholar

🌐 LinkedIn

EXPERIENCE

Researcher & Lecturer - EPFL (CH)

Since 2025

Research and Teaching

Research on privacy-enhancing technologies, such as, synthetic data, privacy-preserving machine learning, and differential privacy. Lectures at BSc and MSc level on information security and privacy.

PhD Research Assistant - EPFL (CH)

2019 – 2024

Research and Teaching

Several high impact publications in the fields of data privacy and trustworthy machine learning.

Data Scientist - Privitar (UK)

2016 – 2019

Product Development, Research, and Consulting

Designed, developed, and prototyped enterprise software that implements privacy-enhancing technologies at scale. Several consulting projects for both public and private sector organisations.

Graduate Student Research Assistant - Werner Reichardt Centre for Integrative Neuroscience (DE)

2015-2016

Experimental Research and Data Analysis

Statistical models of visual information processing in retinal ganglion cells.

Student Research Assistant - University of Erlangen (DE)

2012 - 2014

Experimental Research and Data Analysis

Biophysical modelling of the molecular mechanisms of chronic pain disorders.

EDUCATION

PhD in Computer Science - EPFL (CH), SPRING Lab

2019 - 2024

PhD Thesis: On the Fundamental Limits of Privacy-Enhancing Technologies.

MSc in Computational Neuroscience - University of Tübingen (DE)

2014 - 2016

Lectures in Statistics, Machine Learning, Dynamic Systems, and Neuroscience

BSc in Biomathematics - University of Erlangen (DE)

2011 - 2014

Lectures in Statistics, Linear Algebra, Physics, and Biology

GRANTS & AWARDS

Nominee for the EPFL Doctorate Award - EPFL (CH)

2024

Teaching Assistant Award - EPFL IC (CH)

2021

Graduate Grant - Studienstiftung des Deutschen Volkes (DE)

2011 - 2016

SELECTED INVITED TALKS

Talk	On the Fundamental Limits of Privacy-Enhancing Technologies - Nokia Bell Labs Invited talk at the Responsible AI seminar.	2025
Lecture	On the fundamental limits of privacy-preserving data sharing technologies - Brussels Privacy Hub Lecture at the Summer Academy For Global Privacy Law 2024	2024
Panel	Looking beyond the EU data strategy: Where next for data use and regulation? - CPDP Panel discussion on the future of data use and regulations	2023
Lecture	Synthetic data as a privacy mechanism - A cautionary tale - MIT Invited lecture in the Health Science and Technology Program	2022

SELECTED MEDIA COVERAGE

Podcast	Privacy Engineering - They Talk Tech Podcast Available at frauen-technik.podigee.io	2025
News Article	Warum wollen plötzlich alle Luca? - Eva Wolfangel, Die Zeit Available at zeit.de	2021

News Article	EU privacy experts push a decentralized approach to COVID-19 contacts tracing - <i>TechCrunch</i> Available at techcrunch.com	2020
News Article	Coronavirus apps: the risk of slipping into a surveillance state. - <i>Financial Times</i> Available at ft.com	2020

SELECTED ACADEMIC SERVICE & INVITED REVIEWS

PC Member	Conference on Fairness, Accountability, and Transparency - <i>FAccT</i>	2024
PC Member	Privacy Enhancing Technologies Symposium - <i>PETS</i>	2019-2025
Invited Reviewer	Workshop on Privacy in Machine Learning - <i>NeurIPS'21</i>	2021
External Reviewer	Conference on Computer and Communications Security - <i>CCS'19</i>	2019
Reviewer	Rethinking data and balancing digital power by the <i>Ada Lovelace Institute</i> Report on a future vision for data use and regulation. Available at adalovelaceinstitute.org	2022

SELECTED PUBLICATIONS

2024

T. Stadler, B. Kulynych, N. Papernot, M. Gastpar, and C. Troncoso. The fundamental limits of least-privilege learning. In *Proceedings of the 41th International Conference on Machine Learning (ICML 24)*, 2024

2022

T. Stadler, B. Oprisanu, and C. Troncoso. Synthetic data – Anonymisation Groundhog Day. In *31st USENIX Security Symposium (USENIX Security 22)*, 2022

T. Stadler and C. Troncoso. Why the search for a privacy-preserving data sharing mechanism is failing. *Nature Computational Science*, 2022

C. Troncoso, T. Stadler, D. Bogdanov, E. Bugnion, S. Chatel, C. Cremers, S. Gürses, J.-P. Hubaux, D. Jackson, J. R. Larus, et al. Deploying decentralized, privacy-preserving proximity tracing. *Communications of the ACM*, 2022

2021

T. Stadler, W. Lueks, K. Kohls, and C. Troncoso. Preliminary analysis of potential harms in the luca tracing system. *arXiv preprint arXiv:2103.11958*, 2021

2020

C. Troncoso, M. Payer, J.-P. Hubaux, M. Salathé, J. Larus, E. Bugnion, W. Lueks, T. Stadler, A. Pyrgelis, D. Antonioli, et al. Decentralized privacy-preserving proximity tracing. *arXiv preprint arXiv:2005.12273*, 2020

V. von Wyl, S. Bonhoeffer, E. Bugnion, M. A. Puhon, M. Salathé, T. Stadler, C. Troncoso, E. Vayena, and N. Low. A research agenda for digital proximity tracing apps. *Swiss Medical Weekly*, 2020

M. Salathé, C. L. Althaus, N. Anderegg, D. Antonioli, T. Ballouz, E. Bugnion, S. Capkun, D. Jackson, S.-I. Kim, J. Larus, et al. Early evidence of effectiveness of digital contact tracing for sars-cov-2 in switzerland. *medRxiv*, 2020

PATENTS

2023

J. D. McFALL, C. C. Cabot, T. J. Moran, K. F. P. Guinamard, V. M. Eatwell, B. T. Pickering, P. D. Mellor, T. Stadler, A. Petre, C. A. Smith, et al. Computer-implemented privacy engineering system and method, Nov. 9 2023. US Patent App. 18/349,223

2022

C. C. Cabot, K. F. P. Guinamard, J. D. McFALL, P.-a. Maugis, P. Hector, B. T. Pickering, T. Stadler, J.-a. Tay, and S. Weller. Method or system for querying a sensitive dataset, Sept. 1 2022. US Patent App. 17/618,765